# Sample Paper 07

## Class - 12th Exam - 2025 - 26

## Mathematics (Code-041)

**Time : 3 Hours** **Max. Marks : 80**

**General Instructions :**

**Read the following instructions very carefully and strictly follow them :**

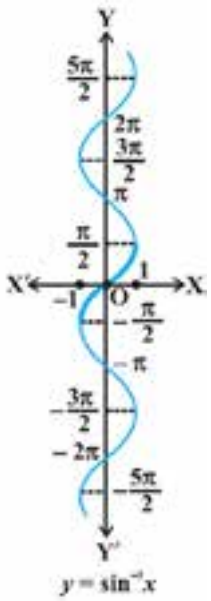
1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

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# SECTION - A

**Section A consists of 20 questions of 1 mark each.**

**1.** At which point does the graph of $y = \sin^{-1} x$ intersect the origin?




(a) (-1,0) (b) (0,0)

(c) (1,0) (d) (0,1)

**2.** The order of the differential equation of all circles of radius $a$ is

(a) 2 (b) 3

(c) 4 (d) 1

**3.** Two dice are thrown $n$ times in succession. The probability of obtaining a doublet six atleast once is

(a) $\left(\frac{1}{36}\right)^n$ (b) $1-\left(\frac{35}{36}\right)^n$

(c) $\left(\frac{1}{12}\right)^n$ (d) None of these

**4.** Solution of $(x+2y^3)\,dy = y\,dx$ is

(a) $x = y^3 + cy$ (b) $x + y^3 = cy$

(c) $y^2 - x = cy$ (d) none of these

**5.** $\int_0^{\pi/2} \frac{\sin x - \cos x}{1+\sin x \cos x}\,dx$ is equal to

(a) 0 (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$ (d) $\pi$

**6.** If the position vectors of the vertices $A, B, C$ of a triangle $ABC$ are $7\hat{j}+10\hat{k}, -\hat{i}+6\hat{j}+6\hat{k}$ and $-4\hat{i}+9\hat{j}+6\hat{k}$ respectively, then triangle is

(a) Equilateral

(b) Isosceles

(c) Scalene

(d) Right angled and isosceles also

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**7.** The derivative of $\log|x|$ is

(a) $\frac{1}{x}$, $x > 0$ (b) $\frac{1}{|x|}$, $x \neq 0$

(c) $\frac{1}{x}$, $x \neq 0$ (d) None of these

**8.** $\cos^{-1}\left[\cos\left(-\frac{\pi}{3}\right)\right]$ is equal to

(a) $-\frac{\pi}{3}$ (b) $\frac{\pi}{3}$

(c) $\frac{2\pi}{3}$ (d) $\frac{5\pi}{3}$

**9.** If $A=\begin{bmatrix}1 & x\\ x^2 & 4y\end{bmatrix}$, $B=\begin{bmatrix}-3 & 1\\ 1 & 0\end{bmatrix}$ and $adj\,(A)+B=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$, then the values of $x$ and $y$ are

(a) 1, 1 (b) $\pm 1$, 1

(c) 1, 0 (d) None of these

**10.** If $f(x)=\begin{cases}ax+3, & x\leq 2\\ a^2x-1, & x>2\end{cases}$, then the values of $a$ for which $f$ is continuous for all $x$ are

(a) 1 and $-2$ (b) 1 and 2

(c) $-1$ and 2 (d) $-1$ and $-2$

**11.** Integrating factor of differential equation $\cos x\frac{dy}{dx}+y\sin x=1$

(a) $\cos x$ (b) $\tan x$

(c) $\sec x$ (d) $\sin x$

**12.** If $\alpha, \beta, \gamma$ are the angles which a half ray makes with the positive directions of the axes, then $\sin^2\alpha+\sin^2\beta+\sin^2\gamma$ is equal to

(a) 2 (b) 1

(c) 0 (d) $-1$

**13.** $\int\frac{1+x+\sqrt{x+x^2}}{\sqrt{x}+\sqrt{1+x}}dx$ is equal to

(a) $\frac{1}{2}\sqrt{1+x}+C$ (b) $\frac{2}{3}(1+x)^{3/2}+C$

(c) $\sqrt{1+x}+C$ (d) $2(1+x)^{3/2}+C$

**14.** If $P(A\cup B)=0.83$, $P(A)=0.3$ and $P(B)=0.6$, then the events will be

(a) dependent

(b) independent

(c) cannot say anything

(d) None of the above

**15.** The function $f(x)=x^3$ has a

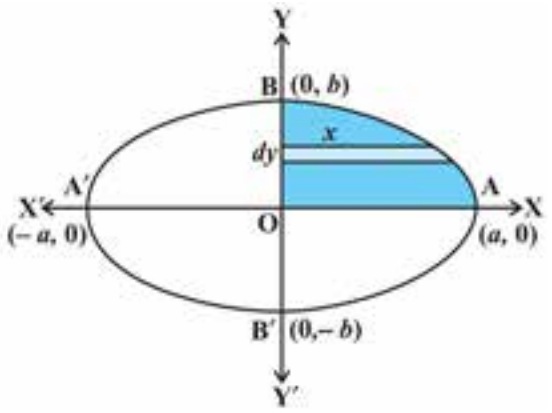
(a) local minima at $x=0$

(b) local maxima at $x=0$

(c) point of inflexion at $x=0$

(d) none of the above

**16.** In Fig., the element marked "dy" in the shaded strip represents:



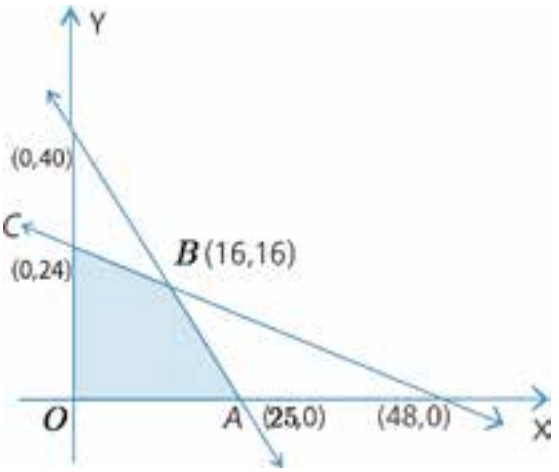

(a) The semi-minor axis of the ellipse

(b) A small change in the y-coordinate (thickness of the strip)

(c) A small change in the x-coordinate (width of the strip)

(d) The total height of the ellipse along the Y-axis

**17.** For the given feasible region in the figure, bounded by the lines shown, consider the objective function $z=2x+y$. The feasible region includes the corner points $O(0,0), A(25,0), B(16,16)$, and $C(0,24)$. At the vertex $B(16,16)$, what is the value of z?




(a) 32 (b) 48

(c) 64 (d) 72

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**18.** The function $f(x) = x^2$, for all real $x$, is

(a) decreasing

(b) increasing

(c) neither decreasing nor increasing

(d) none of the above

**19.** **Assertion:** If $A = \frac{1}{3}\begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ -2 & -2 & -1 \end{bmatrix}$, then $(A^T)\,A = I$

**Reason:** For any square matrix $A$, $(A^T)^T = A$

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

**20.** **Assertion:** Let $A = \{-1,1,2,3\}$ and $B = \{1,4,9\}$ where $f: A \to B$ given by $f(x) = x^2$, then $f$ is a many-one function.

**Reason:** If $x_1 \neq x^2 \Rightarrow f(x_1) \neq f(x_2)$, for every $x_1, x_2 \in$ domain then $f$ is one-one or else many

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

# SECTION - B

**This section comprises of very short answer type-questions (VSA) of 2 marks each.**

**21.** What is the range of the function

$$f(x) = \frac{|x-1|}{x-1},\ x \neq 1?$$

**22.** If $|\vec{a} \times \vec{b}|^2 + |\vec{a} \cdot \vec{b}|^2 = 400$ and $|\vec{a}| = 5$, then write the value of $|\vec{b}|$.

**OR**

If $\vec{a} = 7\hat{i} + \hat{j} - 4\hat{k}$ and $\vec{b} = 2\hat{i} + 6\hat{j} + 3\hat{k}$, then find the projection of $\vec{a}$ on $\vec{b}$.

**23.** Find the general solution of following equation $e^x \tan y\,dx + (1 - e^x)\sec^2 y\,dy = 0$

$$\frac{\tan y}{e^x - 1} = C$$

$$\tan y = C(e^x - 1)$$

$\tan y = C(e^x - 1)$ is the required solution.

**24.** If a line makes angles $90^\circ$, $60^\circ$ and $\theta$ with $X$, $Y$ and $Z$-axis respectively, where $\theta$ is acute angle, then find $\theta$.

**25.** Write the value of $\int \frac{\sec^2 x}{\operatorname{cosec}^2 x}\,dx$.

**OR**

Write the value of $\int \frac{dx}{x^2 + 16}$.

# SECTION - C

**This section comprises of short answer-type questions (SA) of 3 marks each.**

**26.** In the interval $\frac{\pi}{2} < x < \pi$, find the value of $x$ for which the matrix $\begin{bmatrix} 2\sin x & 3 \\ 1 & 2\sin x \end{bmatrix}$ is singular.

**27.** Evaluate $P(A \cup B)$, if $2P(A) = P(B) = \frac{5}{13}$ and $P\left(\frac{A}{B}\right) = \frac{2}{5}$.

**28.** If $e^y(x+1) = 1$, then show that $\frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2$.

**29.** Write the value of $\cos^{-1}\left(\cos\frac{7\pi}{6}\right)$.

**30.** Evaluate $\int_0^\pi \frac{x \sin x}{1 + \cos^2 x}\,dx$.

**OR**

Evaluate $\int_0^\pi \frac{x \tan x}{\sec x + \tan x}\,dx$.

**31.** The sides of an equilateral triangle are increasing at the rate of 2 cm/s. Find the rate at which the area increases, when the side is 10 cm?

**OR**

Find the value(s) of $x$ for which $y = [x(x-2)]^2$ is an increasing function.

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# SECTION - D

**This section comprises of long answer-type questions (LA) of 5 marks each.**

**32.** Evaluate $\int \frac{x \sin^{-1} x}{\sqrt{1-x^2}} dx$.

**33.** Maximise and minimise $Z = x + 2y$ subject to the constraints

$$x + 2y \geq 100$$
$$2x - y \leq 0$$
$$2x + y \leq 200$$
$$x, y \geq 0$$

Solve the above LPP graphically.

**OR**

Maximise $Z = 8x + 9y$ subject to the constraints given below

$$2x + 3y \leq 6$$
$$3x - 2y \leq 6$$
$$y \leq 1$$
$$x, y \geq 0.$$

**34.** Solve the following differential equation.

$$\left[y - x\cos\left(\frac{y}{x}\right)\right]dy + \left[y\cos\left(\frac{y}{x}\right)2x\sin\left(\frac{y}{x}\right)\right]dx = 0$$

**OR**

Find the particular solution of the differential equation $(3xy + y^2)\,dx + (x^2 + xy)\,dy = 0$, for $x = 1$ and $y = 1$.

**35.** Find the shortest distance between the lines

$$\vec{r} = (4\hat{i} - \hat{j}) + \lambda(\hat{i} + 2\hat{j} - 3\hat{k})$$

and $$\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \mu(2\hat{i} + 4\hat{j} - 5\hat{k}).$$

**OR**

Find the shortest distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$.

# SECTION - E

**Case study based questions are compulsory.**

**36.** At its simplest, a fair die states that each of the faces has a similar probability of landing facing up. A standard fair six-sided die, for example, can be regarded as "fair" if each of the faces consists of a probability of 1/6.



A fair die is rolled. Consider the events $A = \{1,3,5\}$, $B = \{2,3\}$, and $C = \{2,3,4,5,\}$

On the basis of above information, answer the following questions.

(i) Find the probability $P(A/B)$ and $P(B/A)$.

(ii) Find the probability $P(A/C)$, $P(A \cap B/C)$ and $P(A \cup B/C)$

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**37.** Sachin Mehara is a final year student of civil engineering at IIT Delhi. As a final year real time project, he has got the job of designing a auditorium for cultural activities purpose. The shape of the floor of the auditorium is rectangular and it has a fixed perimeter, say. $P$.



Based on the above information, answer the following questions.

(i) If $l$ and $b$ represents the length and breadth of the rectangular region, then find the relationship between $l$ ,$b$, $p$.

(ii) Find the area $(A)$ of the floor, as a function of is $l$

(iii) College authority is interested in maximising the area of the floor $A$. For this purpose, find the value of $l$.

**OR**

Find the maximum area of the floor.

**38.** Pfizer Inc. is an American multinational pharmaceutical and biotechnology corporation headquartered on 42nd Street in Manhattan, New York City. The company was established in 1849 in New York by two German immigrants, Charles Pfizer and his cousin Charles F. Erhart. Pfizer develops and produces medicines and vaccines for immunology, oncology, cardiology, endocrinology, and neurology.




The purchase officer of the Pfizer informs the production manger that during the month, following supply of three chemicals, Asprin $(A)$, Caffieine $(C)$ and Decongestant $(D)$ used in the production of three types of pain-killing tablet will be 16, 10 and 16 kg respectively. According to the specification, each strip of 10 tables of Paingo requires 2 gm of $A$, 3 gm of $C$ and 1 gm of $D$. The requirements for other tables are:

| | | | |
|---|---|---|---|
| $X$-prene | 4 gm of $A$ | 1 gm of $C$ | 3 gm of $D$ |
| Relaxo | 1 gm of $A$ | 2 gm of $C$ | 3 gm of $D$ |

(i) Taking the suitable variable form the system of equation that represent given problem.

(ii) Use matrix inversion method to find the number of strips of each type so that raw materials are consumed entirely.

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